

# BoQ -> Digital Material Passport AI Platform

## 2.5-Minute Executive Presentation Script & Demonstration Guide

### [0:00 - 0:25] The Core Problem: Blurry & Scanned Image PDFs

Good morning/afternoon everyone. In construction auditing, real-world Bill of Quantities documents—like this 13-page CBRI Roorkee PDF—are scanned, blurry, image-only files with handwritten annotations and non-selectable text.

Traditional text scrapers fail completely on such documents. To solve this, we built a 3-engine Vision AI platform in partnership with the Google Centre for Climate Technology and the PSA Office, Govt. of India, converting raw scanned image pages into standardized, carbon-quantified Digital Material Passports.

### [0:25 - 0:55] Multi-Engine Vision Architecture (Action: Show Sidebar)

Rather than relying on a single AI model, our platform runs three complementary vision engines simultaneously in Compare Mode:

- 1. Current OCR Engine (Custom PyMuPDF & EasyOCR baseline pipeline)
- 2. OpenAI Vision Engine (GPT-4o / GPT-4o-mini API handler with Vision prompt)
- 3. Google Gemini Vision Engine (Gemini 2.5 Flash / 1.5 Flash API handler with Vision prompt)

All three models analyze spatial page layout and extract raw line items into a strict canonical schema containing item codes, descriptions, handwritten quantities, units, and structural classifications.

### [0:55 - 1:30] Majority Voting & Consensus Matrix (Action: Click Tab 2)

Because AI models are not absolute truth, we built a Field-by-Field Consensus Engine that normalizes dimensions and executes majority voting. In our Engine Comparison Matrix, extractions are color-coded in real time:

- ☒ 3/3 Full Agreement (Green)
- ☐ 2/3 Majority Vote (Yellow)
- ☐ Disagreement / Review (Red)

Red items flag extraction disagreements, instantly routing items to our human review queue without fabricating false certainty.

### [1:30 - 2:05] Auditability, BBox Evidence & Crop Nuances (Action: Click Tab 1 & 3)

Full auditability is crucial for engineering standards. Under Material Passport Dataset, clicking any item crops and renders the exact 300 DPI PDF source region directly from the original document scan.

Now, because scanned PDF pages contain line skew and spatial offset, bounding box crops may occasionally capture an adjacent item line, but they reliably pinpoint the exact source page region for instant verification. If an item is flagged red, engineers inspect the cropped evidence in the Human Review Queue, edit fields, and confirm overrides.

### [2:05 - 2:30] Embodied Carbon Analytics & Deliverables (Action: Click Tab 4 & Sidebar)

Finally, the system computes cradle-to-gate embodied carbon (A1-A3) using ICE Database v3.0 emission factors—quantifying a total of 40.693 tonnes of CO2e across 44 material-bearing items. All data passes 14 programmatic validation rules and exports into standard 50-column Excel templates and JSON material passports with a single click. Thank you!

### PANEL Q&A QUICK REFERENCE CHEAT SHEET

- |                        |                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Q: Scanned/Blurry PDF? | A: Multi-engine vision pipeline (OCR + OpenAI + Gemini) bypasses traditional text scraping limitations. |
| Q: API Key Security?   | A: Loaded via .env or Streamlit Secrets (OPENAI_API_KEY, GEMINI_API_KEY). Never exposed in UI.          |
| Q: BBox Crop Offset?   | A: Bounding box crops pinpoint precise 300 DPI page regions; minor spatial offsets are easily verified. |
| Q: Carbon Baseline?    | A: 40.693 t CO2e (40,693.01 kg) across 44 material items, cited with ICE v3.0 emission factors.         |